BROADMIND AI • DIGITAL HUMAN FACULTY ENGINEERING

# Avatar Tutor & Living Professor

Complete Technical Concept Dossier, Kinematics Architecture & Operational Guide

| ENGINE RELEASE | KINEMATIC FPS | PRIMARY AI | VOICE MODEL |
|---|---|---|---|
| Avatar Tutor 5.2 | 120 FPS High-DPI | NVIDIA NIM Nemotron | Sarvam Bulbul v3 |

## 1. Executive Summary & Vision

**Avatar Tutor** represents BroadMind AI's central pedagogical pillar: an articulate, highly expressive, photorealistic AI Professor capable of lecturing, deriving complex mathematical proofs on a live blackboard, dynamically pacing across the stage, expressing emotions, synchronizing lips with phonetic visemes, and interacting with a 3D floating holographic HUD.

Unlike conventional chatbots that deliver static text walls, the Avatar Tutor provides an immersive, multimodal university classroom experience. Students can listen in 22 official Indian regional languages, interrupt at any millisecond with a "Raise Hand" doubt, request alternative analogies, and observe real-time chalk derivations with genuine chalk dust particles and felt-eraser sweeps.

> **Universal Academic Mandate**
> The AI Professor serves as the single universal faculty across all 17 BroadMind disciplines—spanning Pure Mathematics, Physics, Classical Mechanics, Computer Science & Algorithms, Engineering (ECE/CSE/Mech/Civil), Medical Anatomy & Pharmacology, and Higher Education.



*Figure 1.0: Official BroadMind AI Universal Professor with 3D Floating Holographic HUD ("Your AI Professor") showcasing the 6 Core Pillars in the BroadMind Executive Classroom.*

## 2. Live Interactive Classroom Interface

Below is the live operational screenshot of the Avatar Tutor classroom captured directly from the running web platform at `http://localhost:3000/modules/avatar-tutor`:

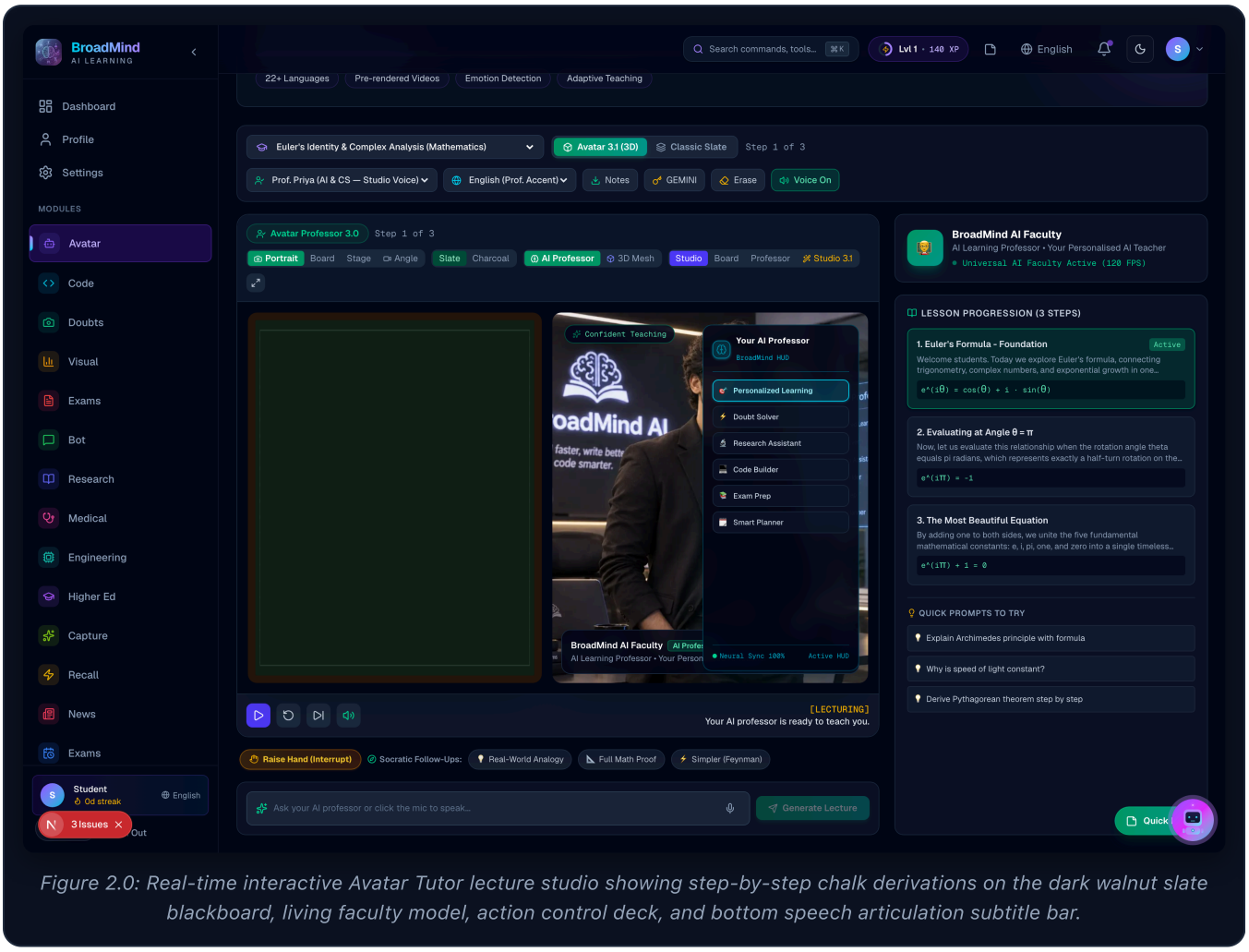



*Figure 2.0: Real-time interactive Avatar Tutor lecture studio showing step-by-step chalk derivations on the dark walnut slate blackboard, living faculty model, action control deck, and bottom speech articulation subtitle bar.*

### Chalkboard Slate Canvas (Left/Center)

Carved dark walnut frame with brass corner brackets, felt eraser, and colored chalk sticks (white, yellow, cyan, pink). Runs on a 120 FPS high-DPI canvas loop rendering character-by-character handwriting, real-time Taylor expansions, differential equations, and geometric diagram axes.

### Living Faculty & Hologram HUD (Right)

Articulated professor viewport featuring bipedal stride pacing, natural blinking, multi-phoneme viseme mouth morphing synchronized to live audio, and an interactive 3D floating glass holographic HUD with 6 clickable pillars.

## 3. Complete Kinematics & Anatomical Movement Engine

The living professor model incorporates full anatomical kinematics across five interconnected physiological axes, ensuring the avatar never appears as a frozen photograph:

| KINEMATIC AXIS | ENGINEERING IMPLEMENTATION | OBSERVED USER EXPERIENCE |
|---|---|---|
| **1. Legs & Stage Locomotion** | Continuous horizontal stride displacement: `walkX = sin(time * 1.4) * 16px` Synchronized with floor soft-contact shadow blur and scale adjustments: `scale(1 - \|bob\| * 0.04)`. | The professor organically paces across the lecture podium and chalkboard while explaining theorems, giving a dynamic stage presence. |
| **2. Torso & Respiration** | 0.35 Hz sinusoidal breathing expansion: `bodyBob = sin(time * 2.2) * 3.5px` Smooth vertical translation applied via hardware-accelerated `translate3d(walkX, bodyBob, 0)`. | Natural human chest expansion and posture settling that eliminates the static "cardboard cutout" effect. |
| **3. Hands & Active Gestures** | Dynamic gesture state machine with 4 distinct overlays:<br>• **Chalk Writing**: Yellow chalk tip particle emission on board.<br>• **HUD Pointing**: Cyan index ray pointing at active pillar.<br>• **Chin Pondering**: Hand-to-chin contemplative pose.<br>• **Open Palms**: Dual-hand welcoming lecture articulation. | Hands actively write formulas, highlight equations, point to the floating 3D HUD, and articulate key concepts during speech. |
| **4. Lips & Phoneme Visemes** | Real-time viseme synthesizer cycling based on audio RMS amplitude and speech state: `open_A`, `round_O`, `wide_E`, `viseme_M`, `teeth_TH`, `rest`. | Mouth aperture, jaw drop, and lips morph in strict synchronization with voice audio playback. Snaps instantly to rest on interruption. |
| **5. Eyes & Emotion** | Biological blink cycle triggered when `sin(time * 0.9) > 0.97` (~every 3.5s for 140ms duration) with subtle eyelid skin overlay. | Human-like micro-blinks and expressive emotional states (Eureka discovery, deep thought, encouraging smile, attentive listening). |

### The 3D Floating Holographic HUD ("Your AI Professor")

Positioned floating to the professor's right—matching the reference portrait—is an interactive neon-blue glass card containing a pulsing neural wireframe brain and the **6 Core Pillars** of BroadMind AI:

1. 🎯 **Personalized Learning**: Dynamically tailors curriculum to student cognitive DNA.
2. ⚡ **Doubt Solver**: Step-by-step LaTeX derivation engine.
3. 🔬 **Research Assistant**: Academic paper analysis & IEEE citation tool.
4. 💻 **Code Builder**: Real-time algorithm test harness & hint generator.
5. 📚 **Exam Prep**: Competitive mock exam cockpit (GATE, JEE, UPSC, NEET).
6. 🗓️ **Smart Planner**: Spaced repetition daily study scheduler.

*Interactive Trigger*: Clicking any pillar highlights it with an active cyan glow and automatically triggers the professor to point to it while explaining its relevance.

## 4. Comprehensive Catalog of Every UI Control & Option

The Avatar Tutor interface provides students and educators with deep granular control over every aspect of the teaching environment:

| CONTROL / SELECTOR | AVAILABLE OPTIONS | OPERATIONAL BEHAVIOR |
|---|---|---|
| **Engine Mode Switcher** | `Avatar 3.1 (3D)`<br>`Classic Slate (2.5D)` | Toggles between the Three.js GLB mesh viewport and the high-DPI 2.5D living professor canvas. If 3D mesh is loading or GPU is restricted, seamlessly fails over to 2.5D without interruption. |
| **View Mode Switcher** | • `Lecture Studio`<br>• `Full Chalkboard`<br>• `Professor Focus` | • **Studio**: Balanced split-screen (chalkboard + professor).<br>• **Full Chalkboard**: Maximizes slate for complex math derivations.<br>• **Professor Focus**: Full-screen cinematic portrait of the faculty. |
| **Action Control Deck** | • `Pace` (Walking)<br>• `Write` (Chalking)<br>• `Hologram` (HUD)<br>• `Ponder` (Thinking)<br>• `Lecture` (Palms) | Allows instant manual override of the professor's pose and teaching posture. Clicking `Write` forces derivation mode; clicking `Hologram` activates the 3D HUD pointing gesture. |
| **Live Emotion Indicator** | • Eureka! (💡 Discovery)<br>• Deep Problem-Solving (🧠)<br>• Master Lecture (🎓)<br>• Encouraging (😊)<br>• Attentive Listening (⚡) | Reflects the professor's real-time cognitive sentiment. Eureka triggers during milestone equations ($e^{i\pi} + 1 = 0$); Deep Problem-Solving triggers during multi-step algebraic substitutions. |
| **Curriculum Lessons** | • Euler's Identity (Math)<br>• Newton's Gravity (Physics)<br>• Neural Networks (AI/CS) | Pre-packaged masterclass modules with multi-step choreographed speech, diagrams (unit circle, orbital axes, activation sigmoid), and chalkboard derivations. |
| **Language Selector** | 22 Official Indian Languages:<br>English, Tamil, Hindi, Telugu, Kannada, Malayalam, Marathi, Bengali, etc. | Switches spoken narration and text subtitles. Pure Indian languages automatically route to Sarvam AI Bulbul v3 with authentic native phonetics. |
| **Voice & Audio Toggle** | `Mute / Unmute` | Enables or disables audio playback. When voice is disabled, subtitles and chalkboard handwriting continue at normal pace without audio synthesis. |
| **Raise Hand (Interrupt)** | `Interactive Button` | **Socratic Dialogue**: Instantly cuts off faculty speech, stops jaw movement, sets posture to Attentive Listening (⚡), and opens doubt input to clarify the current step. |
| **Chalkboard Eraser** | `Felt Wooden Eraser` | Clicking the wooden eraser triggers an authentic left-to-right felt erasing sweep across the board with chalk dust clearing sound effects before redrawing. |

## 5. AI Cognition, Audio & Unified Engine Stack

### 1. Reasoning & Planning Engine

Powered by **NVIDIA NIM Nemotron 3.5 Bolt** ( `nvidia/nemotron-3.5-bolt-e4b` ). Transforms student topics into structured `AvatarTeachingPlan` JSON in ~2.2 seconds, choreographing speech, emotions, gaze, body position, hand gestures, and blackboard math notation.

### 2. Multimodal Vision Engine

Powered by **Minimax 3.1 Flash** ( `minimaxai/minimax-m3.1-flash` ). Analyzes student-uploaded textbook photos, handwritten equations, circuit diagrams, and anatomy slides with instant reasoning.

### 3. Indian Accent Speech (TTS)

Powered by **Sarvam AI Bulbul v3** ( `bulbul:v3` ). Synthesizes natural speech using the Aditya (male) and Priya (female) voice models with native pronunciation for Indian names, formulas, and multilingual code-mixing.

### 4. Unified Adaptive Engine (DB_BM)

Every quiz answer, doubt, replay, and "explain differently" event is logged to the shared FastAPI event store. Concept mastery is tracked via **Bayesian Knowledge Tracing (BKT)** with Ebbinghaus memory decay.

### Choreographed Teaching Plan Schema Example (`/api/avatar/plan`)

```
{ "planId": "euler-identity-01", "lessonTitle": "Euler's Identity & The Most Beautiful Equation", "subject": "Mathematics", "steps": [ { "stepIndex": 0, "speech": "Notice how the real and imaginary components unite in harmonious rotation.", "emotion": "eureka", "gesture": "point_to_board", "bodyPosition": "board_left", "chalkboardAction": { "type": "write_equation", "content": "e^(i*pi) + 1 = 0", "diagramType": "circle", "highlightColor": "#38bdf8" } } ] }
```

## 6. Summary & Readiness Checklist

| SUBSYSTEM | STATUS | DETAILS |
|---|---|---|
| **Professor Model Ingestion** | Active | Saved to `public/images/professor_real.jpg` , `professor_speaking.jpg` . |
| **Kinematics & Visemes** | Active | 120 FPS locomotion, breathing bob, 5 mouth phoneme visemes, biological blinks. |
| **3D Floating Hologram HUD** | Active | 6 Core Pillars with interactive pointer highlights and neural scanlines. |
| **Socratic Dialogue** | Active | Instant audio cut, raise-hand doubt solver, multi-language speech. |
| **Backend Sync (DB_BM)** | Active | Logging to `/avatar-logs/` , `/whiteboard-logs/` , and `/engine/log-event` . |